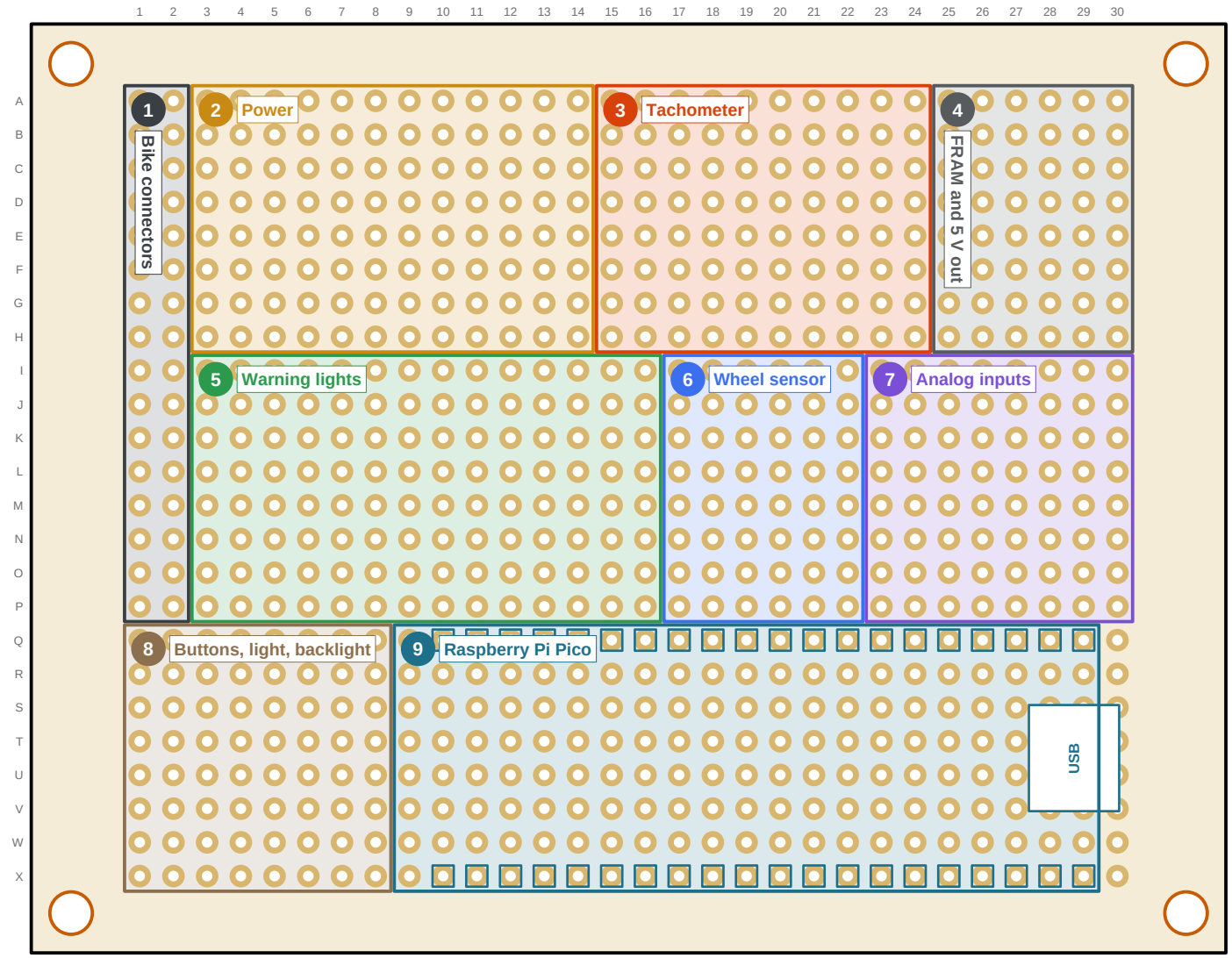


Perfboard 90 × 70 mm, 30 × 24 holes

Seen from the component side: harness connectors on the left edge, Pico USB at the right edge. One zone per schematic sheet.

1 Bike connectors — sheet 1, 4 · 1–2 / A–P
J101 (6 pins), J102 (4 pins): adapter harness
J401 (3 pins): wheel sensor cable

2 Power — sheet 1 · 3–14 / A–H
Q101 IRF4905 lying flat · R101 10k · D101 12 V
D102 SMCJ24A · L101 22 µH · C101 100 µF · C102 1 µF
U101 Pololu D36V28F5 in a socket · D103 1N5819 · C103 470 µF

3 Tachometer — sheet 3 · 15–24 / A–H
D301 1N4007 · R301, R302 22k 1 W, raised off the board
D302 1N4744A · C301 2.2 nF · R303 2.2k
U301 H11L1 in a socket · R304 4.7k · C302 100 nF

4 FRAM and 5 V out — sheet 1, 6 · 25–30 / A–H
U602 FRAM breakout on an 8-pin header
R601, R602 2.2k · C601 100 nF
J103 (2 pins): 5 V to the Pi

5 Warning lights — sheet 2 · 3–16 / I–P
5 channels, each: EL817 in a socket, 10k, 1N4148,
100 nF and 4.7k (3.3k for neutral); outputs to GP6–GP10

6 Wheel sensor — sheet 4 · 17–22 / I–P
U401 EL817 · R401 2.2k · R402 4.7k · D401 1N4148 · C401 100 nF

7 Analog inputs — sheet 5 · 23–30 / I–P
R501 100k · R502 22k · R503 100 Ω · R504, R505 1k
C501–C503 100 nF · D501–D503 BAT54S on SOT-23 adapters

8 Buttons, light, backlight — sheet 5–7 · 1–8 / Q–X
J601 buttons · J501 TEMT6000 · J701 to the TPS61169 breakout
R603, R604 1k · C602, C603 100 nF · R701 1k · R702 100k

9 Raspberry Pi Pico — sheet 6 · 9–29 / Q–X
U601 on 2 × 20 headers: columns 10–29, rows Q and X
USB pair on the TP2/TP3 pads under the Pico (J702)

Print at 100 % (actual size): this bar measures 50 mm

0 5 10 15 20 25 mm

Not on this board

F101 inline fuse (adapter harness) · U701 TPS61169 next to the Kippah · SW601, SW602 and the TEMT6000 on the enclosure · sheet 8, backup LEDs: on the panel (chapter 06)

Mounting holes 84 × 64 mm, from enclosure.scad: [TO VERIFY] on the board in hand.

github.com/frascella-dev/tdm-dash · generated by hardware/drawings/generate.py from hardware/drawings/perfboard.py, do not edit by hand

tdm-dash sensor module · Yamaha TDM 850 (3VD) CERN-OHL-P-2.0

Perfboard layout

SCALE	REV	DATE	SIZE
2:1	0.2	2026-10-21	A4